

Indraprastha Institute of Information Technology, Delhi

IIIT-Delhi - Department of Computer Science & Engineering

Database Management Systems - Course Syllabus

Course Resource Package

Course Code: CSE202 | Course: Database Management Systems

1. Data Models: ER diagrams, Relational model, Relational algebra & calculus.
2. Query Languages: Structured Query Language (SQL), nested queries, integrity constraints.
3. Relational Database Design: Functional dependencies, 1NF, 2NF, 3NF, BCNF, multi-valued dependency.
4. Transaction Processing: ACIDs, serializability, conflict vs view serializability.
5. Concurrency Control: Two-Phase Locking (2PL), timestamp ordering, deadlock handling.
6. Database Internals: Indexing (B-Trees, B+ Trees), hashing, query optimization.